

Vessel Maintenance Manual

M/V Sample Carrier — Engine Department

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Contents

1	Overview and Safety	3
1.1	Scope	3
1.2	General Safety Precautions	3
2	Routine Maintenance Procedures	4
2.1	Fuel System	4
2.1.1	Fuel Filter Element Replacement	4
2.2	Cooling System	5
2.2.1	Centrifugal Cooling Water Pump Overhaul	5
3	Planned Maintenance Schedule	6
4	Troubleshooting Reference	7

1 Overview and Safety

1.1 Scope

This manual describes routine maintenance procedures for the main propulsion plant and supporting auxiliaries of the M/V Sample Carrier. It is intended for qualified engine room staff. All work must be performed in accordance with the vessel's Safety Management System (SMS) and the manufacturer's instructions where applicable.

1.2 General Safety Precautions

Before commencing any maintenance task, the responsible engineer must:

- Confirm the equipment is isolated electrically and mechanically.
- Display a 'Do Not Operate' tag at the local control station.
- Ensure all stored energy (pressure, hydraulic, spring) is released or restrained.
- Wear appropriate personal protective equipment (PPE) for the task.
- Notify the Chief Engineer and the bridge before any operation that affects propulsion or steering.

WARNING: Hot surfaces, pressurised fluids, and rotating machinery are present throughout the engine room. Lockout/Tagout (LOTO) procedures must be followed without exception.

2 Routine Maintenance Procedures

2.1 Fuel System

The fuel system supplies clean, conditioned fuel to the main engine and auxiliary engines. Filter elements should be replaced at the intervals specified in Section 3, or earlier if the differential pressure across the filter housing exceeds 0.8 bar at rated load.

2.1.1 Fuel Filter Element Replacement

Required spare: FL-2200 (Fuel filter element, Mann). Confirm two units are on board before commencing; the duplex filter housing requires one element per side.

Tools: Filter wrench (TL-series), drain tray, lint-free rags, torque wrench.

Estimated time: 45 minutes per side.

Procedure:

1. Inform the bridge and the duty engineer that the fuel filter is being changed over. Reduce engine load to manoeuvring RPM if practicable.
2. Switch the duplex filter changeover lever to the standby (clean) side. Confirm fuel pressure at the engine inlet remains stable; if pressure drops below 4 bar, return the lever to its original position and investigate.
3. Close the inlet and outlet isolation valves of the dirty filter side to be serviced.
4. Place the drain tray under the filter housing. Open the vent screw to relieve pressure, then open the drain valve and drain the housing fully into the tray.
5. Loosen the filter cover bolts evenly in a star pattern and lift the cover off. Inspect the cover gasket; replace if damaged or hardened.
6. Withdraw the used filter element and place it in the waste-oil receptacle. Do not allow the dirty side of the element to contact the clean filter housing.
7. Wipe the inside of the housing clean with a lint-free rag. Inspect the housing for corrosion, cracks, or sediment build-up; report any defects to the Chief Engineer.
8. Lightly coat the gasket of the new FL-2200 element with clean fuel oil. Install the new element, ensuring it seats squarely on the centre spindle.
9. Refit the cover with a new gasket if required. Torque the cover bolts evenly in a star pattern to 35 Nm.
10. Close the drain valve. Open the inlet valve slowly to fill the housing while keeping the vent screw open. Close the vent screw when clean fuel (no air) flows from it.
11. Open the outlet valve fully. Switch the duplex changeover lever back to the dual (both sides in service) position.
12. Run the engine for 10 minutes at normal load and check for leaks at the cover, drain, and vent connections. Record the differential pressure reading and the change-out in the engine log and the Planned Maintenance System.

NOTE: The duplex housing allows one side to be serviced while the other remains in service. Never isolate both sides simultaneously while the engine is running.

2.2 Cooling System

The central cooling system uses a centrifugal pump to circulate fresh water through the main engine jacket cooling, lube oil cooler, and air cooler. Sea-water cooling rejects heat from the central freshwater loop via the central cooler.

2.2.1 Centrifugal Cooling Water Pump Overhaul

Equipment: PM-3105 (Centrifugal cooling water pump, Azcue). Overhaul interval: 16,000 running hours or three years, whichever is sooner. The pump may also be overhauled following any abnormal vibration or performance degradation.

Spares required:

- 1 × mechanical seal kit (matched to PM-3105)
- 2 × bearings (refer to maker's parts list)
- 1 × flange gasket (GK-series)
- 1 × impeller wear ring (if measurement exceeds limits)

Procedure (high-level):

1. Isolate, drain, and tag out the pump in accordance with Section 1.2.
2. Disconnect motor coupling and remove pump casing fasteners. Record alignment readings before separation.
3. Withdraw rotating element. Inspect impeller, wear rings, and shaft for erosion, pitting, or scoring.
4. Replace the mechanical seal kit and bearings. Renew the impeller wear ring if clearance exceeds 0.5 mm.
5. Reassemble with new gaskets. Hand-rotate the shaft to confirm free rotation before refitting the casing.
6. Re-align the coupling within the maker's tolerance (typical 0.05 mm parallel, 0.05 mm angular).
7. Restore services, vent the casing, and run on no-load for 30 minutes while monitoring vibration, bearing temperature, and seal leakage.

3 Planned Maintenance Schedule

The following intervals apply under normal operating conditions. Intervals are stated in main engine running hours unless otherwise specified.

Equipment	Task	Interval	Reference Spare
Main Engine	Fuel filter element replacement	2,000 h	FL-2200
Main Engine	Lube oil filter replacement	1,000 h	FL-series
Main Engine	Air cooler cleaning	4,000 h	—
Main Engine	Cylinder head inspection	8,000 h	GK-series
Aux Engine	Valve clearance check	1,500 h	—
Aux Engine	Fuel injector overhaul	6,000 h	—
Cooling Sys	Centrifugal cooling water pump overhaul	16,000 h	PM-3105
Cooling Sys	Sea water strainer cleaning	1 month	—
Lubrication	Lube oil sample for laboratory analysis	3 months	—
Electrical	Insulation resistance test (main switchboard)	6 months	—
Safety	Emergency generator test run	1 week	—

Where a Reference Spare is given, the listed Item Code corresponds to the vessel's spare-parts inventory. Confirm stock levels in the inventory system before scheduling the work.

4 Troubleshooting Reference

The following table summarises common alarms and the first-line response expected of the watchkeeping engineer. It is not exhaustive; refer to the relevant equipment manual for diagnosis beyond first-line response.

Symptom	Likely Cause	First-Line Action
Fuel filter differential pressure high	Filter element near end of life; contamination	Switch to standby filter, plan FL-2200 replacement
Cooling water pump vibration high	Bearing wear; coupling misalignment	Reduce load, monitor; schedule PM-3105 overhaul
Lube oil inlet temperature high	Cooler fouling; reduced sea-water flow	Check sea water side; back-flush cooler at next opportunity
Auxiliary engine load fluctuating	Governor instability; fuel air entrainment	Bleed fuel system; if persistent, transfer load to standby engine
Exhaust temperature deviation, single cylinder	Injector defect; valve clearance out of spec	Reduce load on cylinder; inspect at first opportunity

— End of Manual —